

EDA and Machine Learning Model Report

1. Dataset Overview

The dataset used is the California Housing dataset. It contains information such as median income, house age, population, and geographic details, with the target variable being median house value (MedHouseVal). Initial exploration showed no missing values and consistent data types.

2. Data Cleaning and Feature Engineering

Duplicates were removed. New features were created including Rooms_per_Household, Bedrooms_per_Room, and Population_per_Household. Irrelevant or redundant columns such as AveRooms, AveBedrms, Population, and Longitude were dropped. Outliers were handled using IQR capping and filtering target values.

3. Exploratory Data Analysis (EDA)

Histograms were plotted to understand distributions. A correlation heatmap revealed strong relationships, especially between median income and house value. Boxplots were used to visualize outliers. Feature engineering improved interpretability and relationships.

4. Model Building

The dataset was split into training and testing sets (80-20). Standardization was applied. A Linear Regression model was trained first, followed by a Random Forest Regressor for comparison.

5. Model Evaluation

Metrics used include Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. Linear Regression achieved moderate performance, while Random Forest showed improved R^2 score (~ 0.73), indicating better prediction capability.

6. Visualization

Scatter plots comparing actual vs predicted values showed that Random Forest predictions aligned more closely with actual values compared to Linear Regression. Feature importance from Random Forest highlighted median income as a key predictor.

7. Improvements

Future improvements include hyperparameter tuning of Random Forest, trying advanced models like Gradient Boosting or XGBoost, performing cross-validation, and adding more domain-specific features. Handling skewness and further feature selection could also improve accuracy.